

Mission Statement:

The primary aim of my project is to explore the intersection of biomedical engineering and biology by researching and diving into various topics. I am currently mentoring under Precigenetics, a company focusing on drug discovery through the application of real-time epigenetic data analysis. Through this project, I hope to understand how engineering and biology converge to form innovative biomedical solutions, and determine whether this field aligns with my professional interests.

Introduction to Original Work:

My original work is divided into three parts, each contributing to different aspects of biomedical engineering, as outlined below:

1. 3D Prototype Design:

- In collaboration with Precigenetics, I created a prototype of the device they are working on. The company provided me with a basic design, and I was tasked with translating that design into a 3D model using CAD software (Fusion360).
- **Importance:** This phase is essential because prototyping in digital space helps engineers visualize the product before committing to physical materials, reducing the cost and resources required for early-stage design testing.

- **Relevance to Engineering:** This task allowed me to apply engineering design principles and CAD software to build a functional 3D model.

2. FDA Regulatory Document Summary:

- In healthcare-related product development, FDA regulations are crucial for ensuring safety and efficacy. However, these regulations are often lengthy and complex. I synthesized a comprehensive 16-page document that condenses essential FDA guidelines for medical devices.
- **Topics Covered:** The document covers device classification, human factors and usability engineering, biocompatibility, material selection, and more, offering clear insights into the FDA approval process for medical devices.
- **Relevance to Engineering:** Understanding regulatory requirements is fundamental for engineers working on medical devices to ensure compliance and successful market entry.

3. Spectroscopy Viewer:

- A key part of understanding biological data, especially for drug discovery, involves spectroscopy. I developed a spectroscopy viewer that expands upon the RGB channels used in basic imaging. By using full-spectrum data, the viewer provides more detailed information on the material composition of biological samples.

- **Relevance to Engineering and Biology:** This tool aids in the analysis of cell structures and can contribute to better understanding of biomarkers or drug interactions in the body.

Project Objectives and Goals:

- **Objective:** The goal of this project is to explore how biomedical engineering integrates with biological sciences. By working on various components like prototyping, FDA regulations, and spectroscopy, I aim to gain hands-on experience in biomedical applications, particularly in drug discovery.
- **Specific Goals:**
 - Develop proficiency in 3D modeling and prototype design for biomedical applications.
 - Understand the regulatory landscape for healthcare products and the steps needed for market approval.
 - Learn and apply advanced spectroscopy techniques to analyze biological data.

Project Methodology:

1. For Prototype Design:

- I began by teaching myself CAD through Fusion360, building up my skills with smaller projects. With prior experience in Maya, transitioning to Fusion360 was relatively smooth. Afterward, I applied these skills to create the 3D model based on Precigenetics' specifications.

2. For FDA Research Paper:

- I reviewed several FDA documents and selected the most relevant sections for my research. After analyzing these documents, I condensed the material into a 16-page overview, summarizing key topics such as device classification, biocompatibility, and ergonomics.

3. For Spectroscopy Viewer:

- I refreshed my Python programming skills to build the viewer. The project required coding and working with spectral data, which was both a challenge and an opportunity to refine my coding abilities. I studied various spectroscopy techniques and applied them in the development of the viewer.

Materials:

All work was conducted using freely available software:

- Fusion360 (Free for students)
- Python (Free open-source tool)
- FDA Regulatory Documents (Available online)
- Spectroscopy Data (Publicly available datasets)

Timeline:

Date	Task
01/24	Begin learning Fusion360 and basic CAD techniques
01/27	Sketch out the design for the prototype based on Precigenetics' specifications
02/01	Start writing the research paper on FDA regulations
02/15	Complete the 3D model of the prototype
03/01	Begin the development of the spectroscopy viewer (Python)
03/10	Finalize the research paper on FDA regulations

Date	Task
03/20	Refine the spectroscopy viewer with additional features
03/31	Begin writing the final report on all components
04/10	Finalize the report, incorporating feedback

Conclusion:

This project has provided a rich learning experience by allowing me to apply engineering principles to real-world problems, particularly in the field of biomedical engineering. By collaborating with Precigenetics and working on components like prototype design, regulatory documentation, and spectroscopy, I have gained a deeper understanding of the complexities and intersections between engineering and biology. This project has helped me determine whether I want to pursue a future in biomedical engineering and drug discovery, and has significantly developed my technical, research, and problem-solving skills.